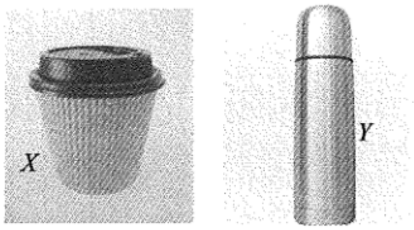
Q1

1.

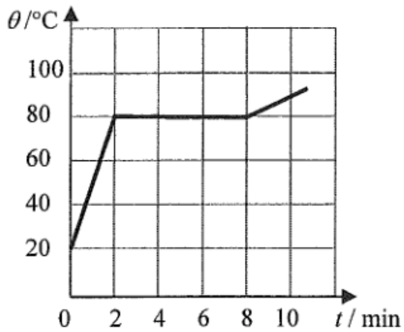


Two identical scoops of ice-cream are transferred from a refrigerator into paper cup X and vacuum flask Y shown above. Under room temperature, the time required for the ice-cream in the containers to melt completely is t_X and t_Y respectively. What is the expected result and explanation ?

A. $t_X > t_Y$ as the vacuum flask reduces heat loss to the surroundings.
B. $t_X > t_Y$ as the vacuum flask retains the heat.
C. $t_Y > t_X$ as the vacuum flask keeps things cold by releasing heat into the surroundings.
D. $t_Y > t_X$ as the vacuum flask reduces the rate of heat gain from the surroundings.

Q2

2.

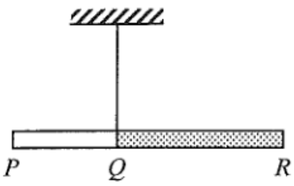


An electric heater of constant power is used to heat a solid substance X which is insulated from the surroundings. The variation of its temperature θ with time t is shown above. X has a specific heat capacity of 800 J kg^{-1} °C^{-1} in its solid state. What is the specific latent heat of fusion of X ?

A. 144 kJ kg^{-1}
B. 192 kJ kg^{-1}
C. 202 kJ kg^{-1}
D. Answer cannot be found as both the mass of X and the power of the heater are not known.

Q3

3.

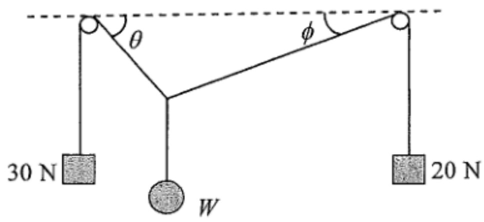


PQR is a composite rod having a uniform cross-section but with portions PQ and QR made of different materials, each of uniform density. The ratio of the length of PQ to that of QR is 2 : 3. When the rod is suspended at Q, it remains horizontal as shown. What is the ratio of the mass of PQ to that of QR ?

A. 2 : 3
B. 1 : 1
C. 3 : 2
D. Answer cannot be found as the ratio of their densities is not given.

Q4

4.



The figure shows a weight W attached to two light strings which pass over two smooth pegs at the same height and with weights 30 N and 20 N attached to the respective ends of each string. The system is at equilibrium. Which deduction about W is correct ?

A. W is less than 50 N.
B. W is equal to 50 N.
C. W is greater than 50 N.
D. No information about W can be obtained as angles θ and ϕ are not known.

Q5

5. A particle is moving along a straight line with uniform acceleration. It takes 4 s to travel a distance of 36 m and then 2 s to travel the next 36 m. What is its acceleration ?

A. 2.5 m s^{-2}
B. 3.0 m s^{-2}
C. 4.0 m s^{-2}
D. 4.5 m s^{-2}

Q6

6. Two small identical blocks slide down from rest on smooth incline planes from the same height H as shown in Figure (1) and Figure (2) below. Their respective speeds at the bottom of the incline planes are denoted by v_1 and v_2 and the respective times taken to reach the bottom are t_1 and t_2. Which of the following is correct ? Neglect air resistance.

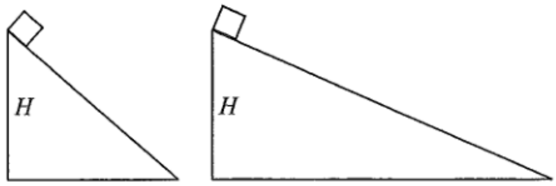
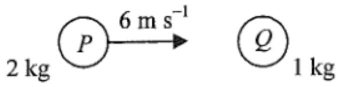


Figure (1) Figure (2)

A. $v_1 > v_2$ and $t_1 = t_2$
B. $v_1 > v_2$ and $t_1 < t_2$
C. $v_1 = v_2$ and $t_1 = t_2$
D. $v_1 = v_2$ and $t_1 < t_2$

Q7

7.



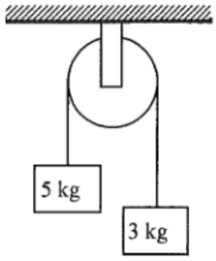
A sphere P of mass 2 kg makes a head-on collision with another sphere Q of mass 1 kg which is initially at rest. The speed of P just before collision is 6 m s^{-1}. If the two spheres move in the same direction after collision, which of the following could be the speed(s) of Q just after collision ?

(1) 2 m s^{-1}
(2) 4 m s^{-1}
(3) 6 m s^{-1}

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

Q8

8. Two blocks of masses 5 kg and 3 kg respectively are connected by a light string passing through a frictionless fixed light pulley. Find the magnitude of the acceleration of the blocks in terms of the acceleration due to gravity g when they are released. Neglect air resistance.

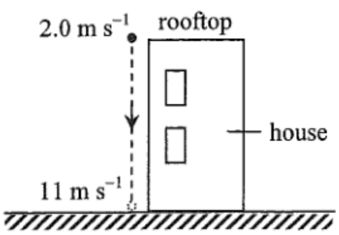


A. g

B. $\frac{g}{\text{[illegible]}}$

Q9

9.

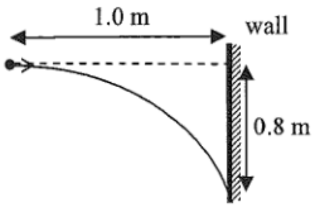


A particle is projected vertically downward with an initial speed of 2.0 m s^{-1} from the rooftop of a house. The particle reaches the ground with a speed of 11 m s^{-1} as shown. Estimate the height of the house. Neglect air resistance. ($g = 9.81$ m s^{-2})

A. 3.3 m
B. 6.0 m
C. 6.5 m
D. 12 m

Q10

*10.



A particle is projected horizontally towards a vertical wall 1.0 m away. It hits the wall at a position 0.8 m vertically below its point of projection. At what speed is it projected ? Neglect air resistance. ($g = 9.81$ m s^{-2})

A. 2.0 m s^{-1}
B. 2.5 m s^{-1}
C. 5.0 m s^{-1}
D. 6.3 m s^{-1}

Q11

*11. An astronaut inside a spacecraft moving in a circular orbit around the Earth is apparently weightless because

A. the astronaut is too far from the Earth to feel the Earth's gravitational force.
B. the astronaut and the spacecraft are both moving with the same acceleration towards the Earth.
C. the Earth's gravitational force on the astronaut is balanced by the reaction force of the spacecraft's floor.
D. the Earth's gravitational force on the astronaut is balanced by the centripetal force.

Q12

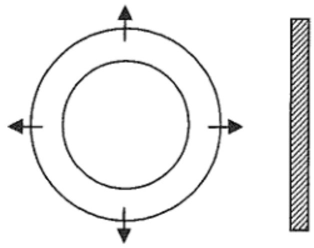
*12. An artificial satellite revolves in a circular orbit above the Earth's surface at a height equal to the radius of the Earth. Find the acceleration of the satellite in terms of the acceleration due to gravity g on the Earth's surface.

A. $\frac{g}{8}$

B. $\frac{g}{4}$

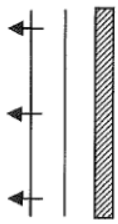
Q13

13.

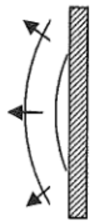


The above figure shows two circular pulses produced by drops of water falling on a ripple tank. The pulses are then reflected by a straight barrier. Which diagram best shows the reflected pulses ?

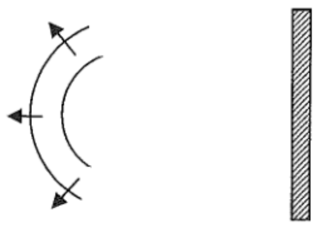
A.



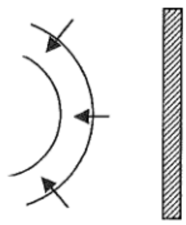
B.



C.

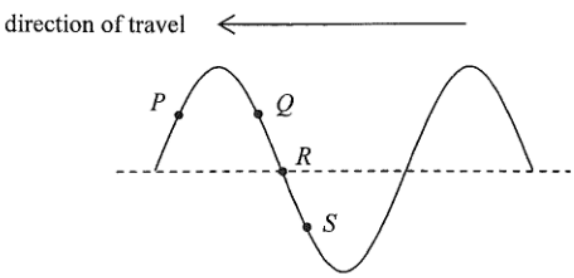


D.



Q14

14. A transverse wave travels towards the left on a long string. *P*, *Q*, *R* and *S* are particles on the string. Which of the following statements correctly describe(s) their motions at the instant shown ?

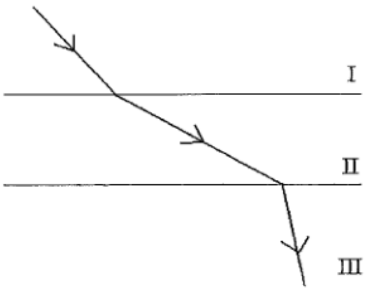


(1) *P* is moving upwards.
(2) *Q* and *S* are moving in opposite directions.
(3) *R* is momentarily at rest.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

Q15

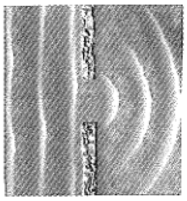
15. The figure shows the path of a light ray travelling from medium I to medium III separated by parallel boundaries. Arrange in **ascending order** the speed of light in the respective media.



A. I < III < II
B. II < III < I
C. III < I < II
D. III < II < I

Q16

16.



The photograph shows a series of plane sea waves travelling through a gap in a sea wall which exhibits diffraction. Assuming that the frequency of the waves remains unchanged, which of the following will increase the degree of diffraction ?

(1) The gap in the sea wall becomes narrower.
(2) The wavelength of the waves increases.
(3) The amplitude of the waves becomes larger.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

Q17

17.

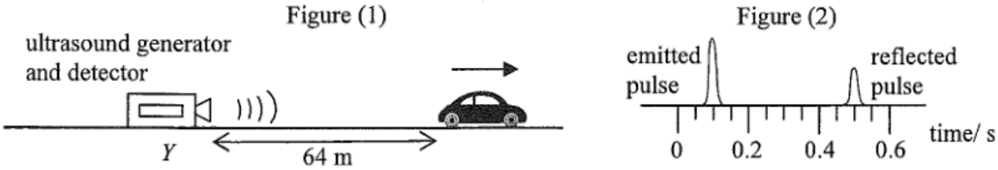
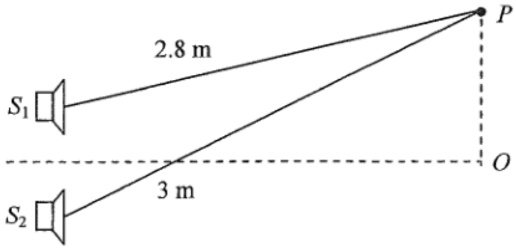


Figure (1) shows a car travelling with a uniform speed along a straight road away from a stationary ultrasound generator and detector at Y. When the car is 64 m from Y, the generator emits an ultrasound pulse towards the car. The pulse is then reflected back to the detector at Y and displayed on a CRO as shown in Figure (2). Estimate the speed of the car. Given: speed of ultrasound in air is 340 m s^{-1}

A. 16 m s^{-1}
B. 20 m s^{-1}
C. 24 m s^{-1}
D. 32 m s^{-1}

Q18

18.



S_1 and S_2 are two loudspeakers connected to a signal generator but the sound waves produced by them are in anti-phase. Point O is equidistant from the loudspeakers while point P is at the distances shown in the figure from the loudspeakers. What type of interference occurs at O and P if the wavelength of the sound waves is 10 cm ?

	O	***P***
A.	destructive	constructive
B.	constructive	constructive
C.	destructive	destructive
D.	constructive	destructive

Q19

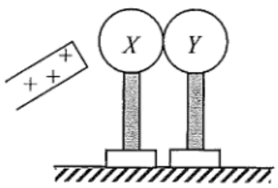
19. Which of the following statements about sound waves is/are correct ?

(1) Sound waves are electromagnetic waves.
(2) Sound waves cannot travel in a vacuum.
(3) Sound waves cannot form stationary waves.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

Q20

20.

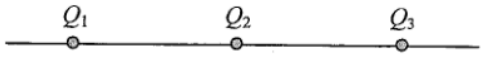


Two insulated uncharged metal spheres X and Y are placed in contact. A positively-charged rod is brought near X as shown. X is then touched by a finger momentarily and the two spheres are then separated by removing Y. The charged rod is removed afterwards. Which of the following describes the charges on X and Y ?

	sphere *X*	**sphere *Y***
A.	uncharged	uncharged
B.	uncharged	positive
C.	negative	uncharged
D.	negative	negative

Q21

21.

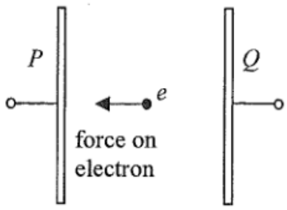


Three point charges Q_1, Q_2 and Q_3 are fixed on a straight line with Q_2 at the mid-point of Q_1 and Q_3. The resultant electrostatic force on each charge is zero. Which of the following can be the sign and magnitude (in the same arbitrary units) of Q_1, Q_2 and Q_3 ?

	Q_1	Q_2	Q_3
A.	+2	+1	+2
B.	+2	−1	+2
C.	−4	+1	+4
D.	−4	+1	−4

Q22

22.

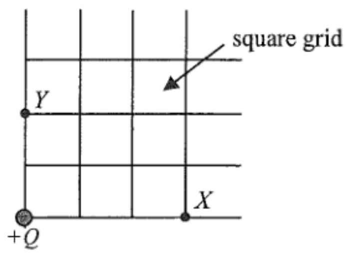


Two parallel metal plates P and Q are maintained at a certain p.d. by a battery (not shown in the figure). An electron placed between the plates would experience an electrostatic force of 8.0×10^{-18} N towards P. Which of the following descriptions about the electric field E between the plates is correct ?

A. $E = 0.02\ \mathrm{N\,C^{-1}}$ from Q to P.
B. $E = 0.02\ \mathrm{N\,C^{-1}}$ from P to Q.
C. $E = 50\ \mathrm{N\,C^{-1}}$ from Q to P.
D. $E = 50\ \mathrm{N\,C^{-1}}$ from P to Q.

Q23

*23.

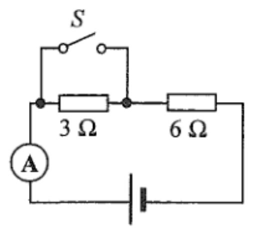


The figure shows the location of an isolated point charge $+Q$. If the electric potential at X is V, what is the electric potential at Y ?

A. $\frac{2}{\text{[illegible]}}V$

Q24

24.

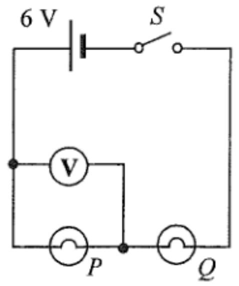


In the above circuit, the cell has constant e.m.f. and a fixed internal resistance. When S is closed, the ammeter reads 3.0 A. When S is open, which of the following is a possible reading of the ammeter ?

A. 1.6 A
B. 2.0 A
C. 2.4 A
D. 3.2 A

Q25

25.

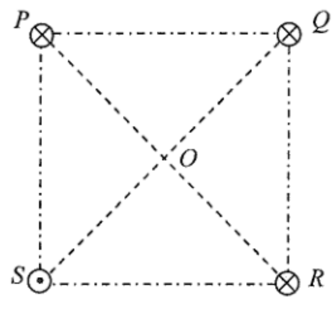


The figure shows two light bulbs P and Q connected to a cell of e.m.f. 6 V and negligible internal resistance. The voltmeter reads 6 V when the switch S is closed. Which of the following is possible ?

A. Both P and Q are short-circuited.
B. Both P and Q are burnt out and become open circuit.
C. P is short-circuited or Q is burnt out and becomes open circuit.
D. P is burnt out and becomes open circuit or Q is short-circuited.

Q26

26.



Four long straight parallel wires P, Q, R and S carrying currents of equal magnitude are situated at the vertices of a square as shown. P, Q and R each carries a current directed into the paper while S carries a current directed out of the paper. The direction of the resultant magnetic field at the centre O of the square is along

A. OP.
B. OQ.
C. OR.
D. OS.

Q27

27.

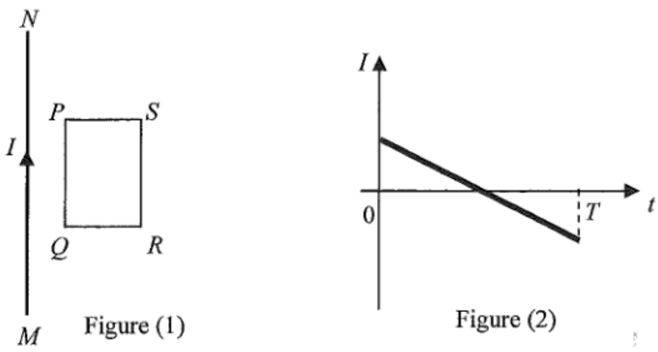


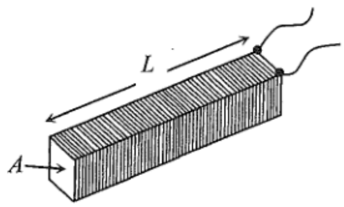
Figure (1) Figure (2)

A long straight current-carrying wire MN and a rectangular coil $PQRS$ are fixed in the same plane as shown in Figure (1). The current I is taken as positive when it flows from M to N and it varies with time t as shown in Figure (2). The direction of the induced current in the coil during the time interval $0 - T$ is

A. first anti-clockwise and then clockwise.
B. first clockwise and then anti-clockwise.
C. anti-clockwise throughout.
D. clockwise throughout.

Q28

28.

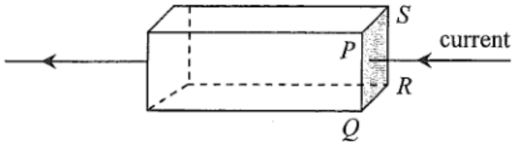


The figure shows a closely packed long solenoid of cross-sectional area A and length L having a total of N turns. If the solenoid carries a constant direct current throughout, which of the following changes can increase the magnetic flux density B at its central cross-section ?

	length	**cross-sectional area**	**total number of turns**
A.	$2L$	$2A$	$2N$
B.	L	$2A$	N
C.	$2L$	A	N
D.	L	A	$2N$

Q29

*29.



The figure shows a current flowing from right to left in a metal block with cross-section $PQRS$. When a uniform magnetic field is applied to the block, the electric potential of the side PQ is found to be higher than that of the side SR. In which direction can the magnetic field be applied ?

A. from P to Q
B. from Q to P
C. from P to S
D. from S to P

Q30

*30. When a heater is connected to a d.c. voltage of 10 V, the power dissipated is P. If the heater is connected to a sinusoidal a.c., the power dissipated becomes $\frac{1}{2}P$. What is the **r.m.s. voltage** of this a.c. source ? Assume that the resistance of the heater is constant.

A. 5 V
B. $5\sqrt{2}$ V
C. 10 V
D. $10\sqrt{2}$ V

Q31

31. Nucleus W decays to nucleus Z as shown below:

$$W \xrightarrow{\alpha} X \xrightarrow{\beta} Y \xrightarrow{\beta} Z$$

Which of the following statements is/are correct ?

(1) Nucleus X has 1 more proton than nucleus Y.
(2) Nucleus W has 2 more neutrons than nucleus X.
(3) W and Z are isotopes of the same element.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

Q32

32. A GM counter is placed close to and in front of a radioactive source which emits both α and γ radiations. The count rate recorded is 450 counts per minute while the background count rate is 50 counts per minute. Three different materials are placed in turn between the source and the counter. The following results are obtained.

Material	**Recorded count rate / counts per minute**
(Nil)	450
cardboard	x
1 mm of aluminium	y
2 mm of lead	z

Which of the following is the most suitable set of values for x, y and z ?

	x	y	z
A.	300	300	100
B.	300	100	50
C.	100	100	0
D.	100	50	50

*33. A radium nucleus decays to a radon nucleus by emitting an α-particle. The energy released in the process is 4.9 MeV. Compared to the mass of a radium nucleus, the total mass of a radon nucleus and an α-particle is

A. 5.4×10^{-11} kg less.
B. 5.4×10^{-11} kg more.
C. 8.7×10^{-30} kg less.
D. 8.7×10^{-30} kg more.